

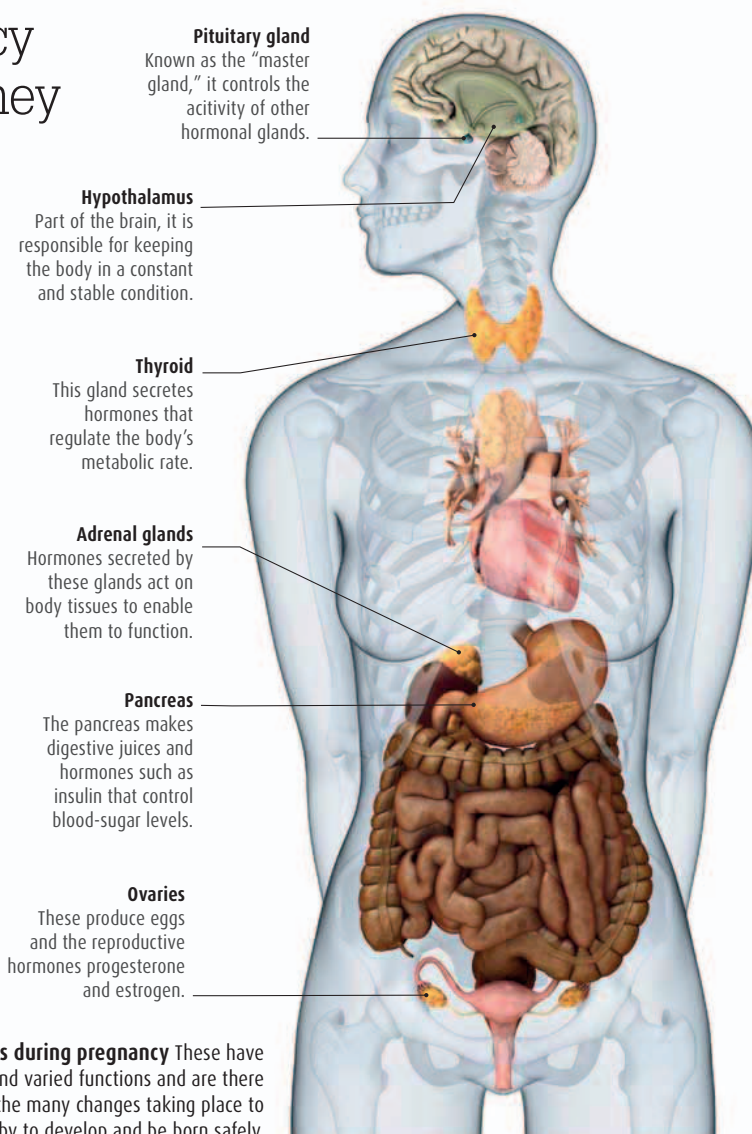
Q I've heard about pregnancy hormones, but what are they and what do they do?

At the beginning of your pregnancy, there is a surge of two hormones, estrogen and progesterone, which have an essential role in maintaining pregnancy. The rise of these hormones is the result of human chorionic gonadotrophin (hCG) signaling to the corpus luteum (the ruptured egg follicle) that pregnancy is underway.

At around six to nine weeks of pregnancy, the placenta takes over the production of estrogen and progesterone for the rest of your pregnancy. Other key hormones during your pregnancy include relaxin, oxytocin, prolactin, and endorphins. These, along with androgens (the male sex hormones), are collectively known as sex steroid hormones. The right balance of hormones is essential for a successful pregnancy.

The body system that regulates all of your hormones is called the endocrine system. This includes various glands and organs throughout your body, including the thyroid, adrenal, and pituitary glands, and the hypothalamus, the pancreas, and the ovaries. All your body systems are affected by the action of your endocrine system.

hCG levels almost double every 48 hours at the beginning of your pregnancy.



ESTROGEN

Initially secreted by the ovaries, estrogen prepares the lining of the uterus for a potential pregnancy. It also has the following key roles during your pregnancy:

- » **Ensures the developing embryo** remains embedded in your uterus.
- » **Dilates your blood vessels** so that blood can flow more easily and in greater quantity.
- » **Stimulates the milk glands** in preparation for breast-feeding.
- » **Improves bone health** while your body is under more pressure.

PROGESTERONE

This hormone plays a vital role in the early stages of pregnancy by preparing the body for fertilization and then establishing the placenta. It does the following:

- » **Prepares the lining of the uterus** for implantation and encourages it to thicken.
- » **Helps to regulate** the changes in your metabolism.
- » **Thickens the "plug"** that forms in the cervix to keep bacteria from entering the uterus and causing infection.
- » **Strengthens muscles** in the pelvic wall in preparation for labor.

OTHER HORMONES

Your pregnancy is also specifically influenced by the action of:

- » **Oxytocin**, the "love" hormone that triggers labor and helps you to bond with your baby.
- » **Endorphins**, the "feel-good" hormones that help to mitigate your experience of pain during labor.
- » **Prolactin**, the hormone that stimulates your body to produce milk for your baby.
- » **Relaxin**, the hormone that helps to loosen your muscles and ligaments to make space for your baby.